

Multi-Camera Multi-Object Tracking

C6 - Video Analysis



Team 5

Álvaro Díaz, Bernat Medina,
Maiol Sabater and Gerard Vilaplana

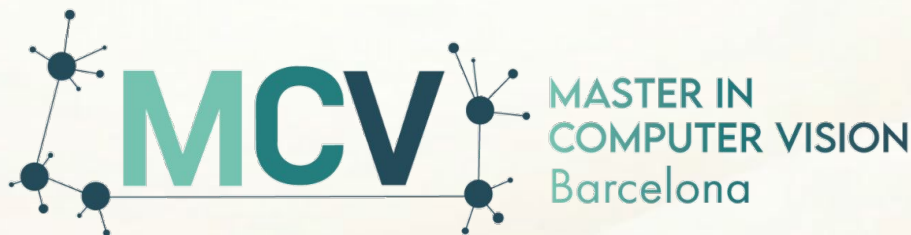


Table of Contents

1	Past Weeks	3
2	The Dataset.....	4
3	Failed Experiments.....	5
4	Our Approach	6
4.1	Detector	7
4.2	Tracker	8
4.3	Re-Identification	9
4.3.1	ReID Logic.....	10-11
5	Results.....	12-18
6	Conclusions	19

Past Weeks

Week 1

- **Adaptive Formulation:**
 - Using learning rate (ρ) to absorb gradual illumination shifts.
- **Bayesian Hyperparameter Search:**
 - Optimizing params and morphological kernels.

Best mAP
(adaptive gaussian)

0.688

Week 2

- **Domain-Specific Fine-Tuning:**
 - Adapted to the specific camera perspectives of our dataset.

Best mAP
(Fine-tuned Faster R-CNN)

0.977

Best IDF1
(Kalman tracker)

0.924

Week 3

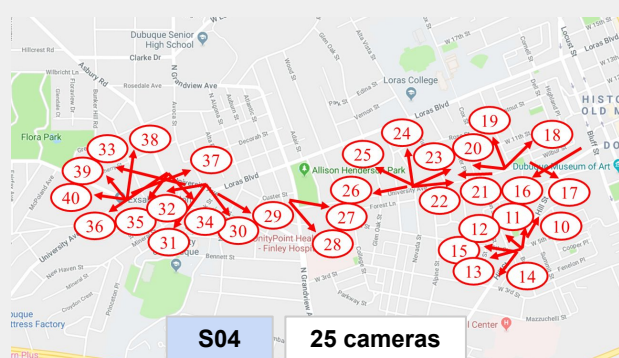
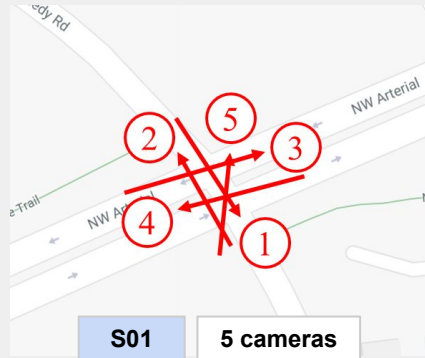
- **Dense Flow Superiority:**
 - Upgrading from classical PyFlow to NeuFlow v2 slashed pixel error (PEPN) from 8.27% to 3.48% and accelerated inference by over 150x.
- **Flow-Based Tracking:**
 - Optimizing params and morphological kernels.

Best IDF1
(NeuFlow tracker)

0.234

The Dataset

Train



Validation



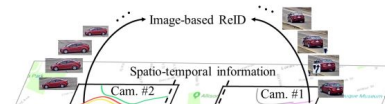
Original Paper

CityFlow: A City-Scale Benchmark for Multi-Target Multi-Camera Vehicle Tracking and Re-Identification

Zheng Tang^{1*} Milind Naphade² Ming-Yu Liu² Xiaodong Yang² Stan Birchfield²
 Shuo Wang² Ratnesh Kumar² David Anastasiu³ Jenq-Neng Hwang¹
¹University of Washington ²NVIDIA ³San Jose State University

Abstract

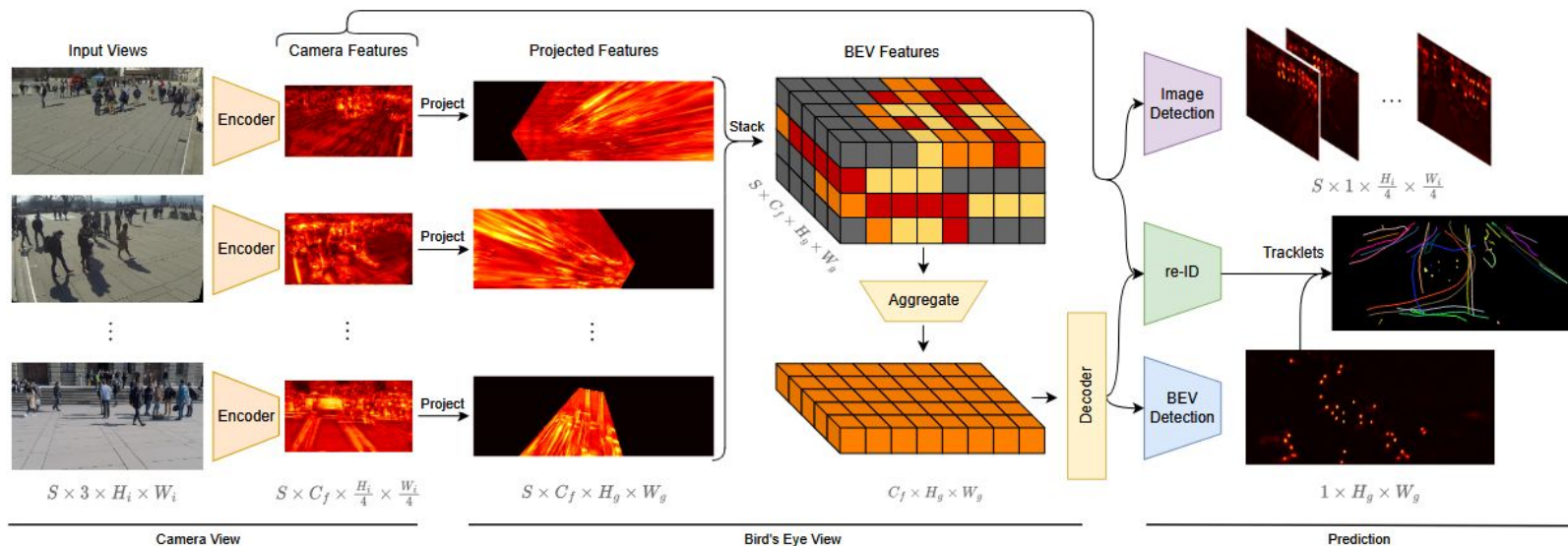
Urban traffic optimization using traffic cameras as sensors is driving the need to advance state-of-the-art multi-target multi-camera (MTMC) tracking. This work intro-



Z. Tang et al., "CityFlow: A City-Scale Benchmark for Multi-Target Multi-Camera Vehicle Tracking and Re-Identification," 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, USA, 2019, pp. 8794-8803.

Failed Experiments

Early Bird



T. Teepe et al., "EarlyBird: Early-Fusion for Multi-View Tracking in the Bird's Eye View," 2024 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW), Waikoloa, HI, USA, 2024, pp. 814-823.

Our Approach

Detection



Yolo26

Motion Tracking

Time: $t-1$...



Time: t



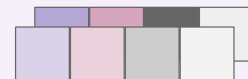
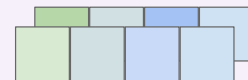
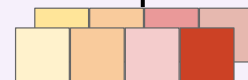
Optical Flow

ReID



ViT

Maiol Sabater



Cosine Sim

ReID Logic

Different ID



Same ID



Detector



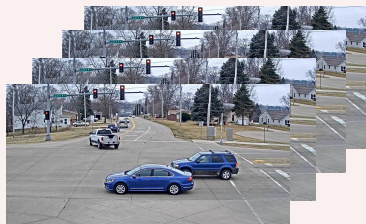
YOLO26

- Fast
- Recent
- Robust



Efficient inference
Latest YOLO version
Strong adaptability

Pipeline



Yolo26



Metrics

mAP50-95	0.54	mAP75	0.65
mAP50	0.66	Speed on CPU	38.9 ms

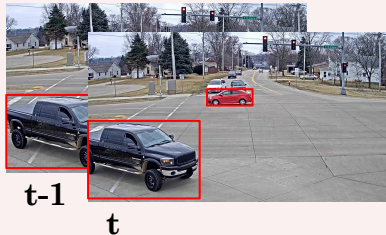


Low mAP
Due: Parked cars

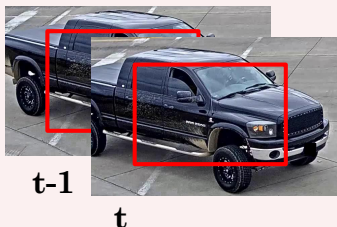


Filter them afterwards

Detection

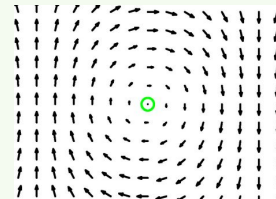


Object detection via YOLO

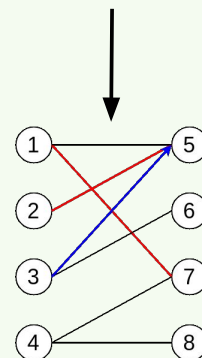


50% inner crop

Motion



Optical flow calculated by NeuFlow v2



Hungarian matching with IoU predictions

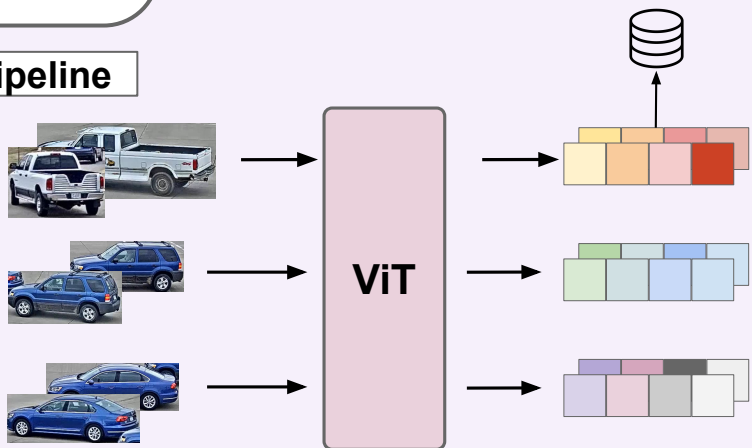
Tracking

$$B_{\text{final}} = \alpha B_{\text{pred}} + (1 - \alpha) B_{\text{det}}$$

State Update on frame t

ReID

Pipeline

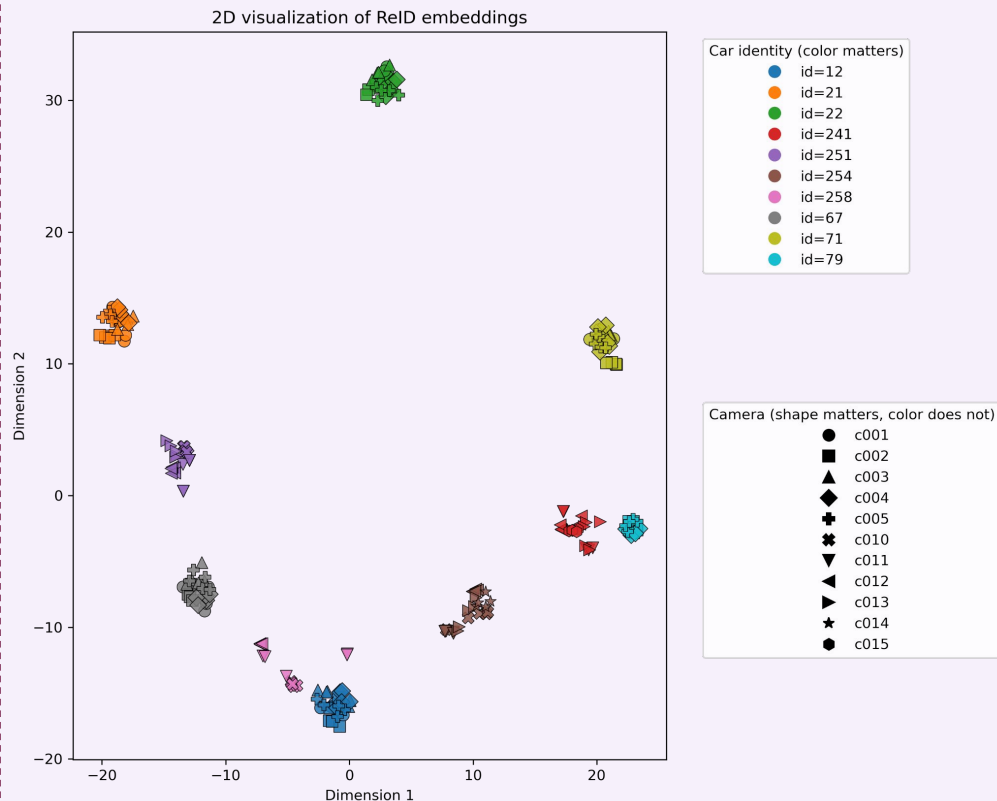


Metrics

Train Loss	0.56	Triplet Loss	0.0021
Val Rank1	0.98	Val mAP	0.68

Triplet loss ≈ 0 $\rightarrow$ embedding space is well separated
Rank-1 $\approx 99\%$ $\rightarrow$ excellent retrieval for top match
mAP $\approx 67\%$ $\rightarrow$ room for improvement in full ranking

Qualitative Results

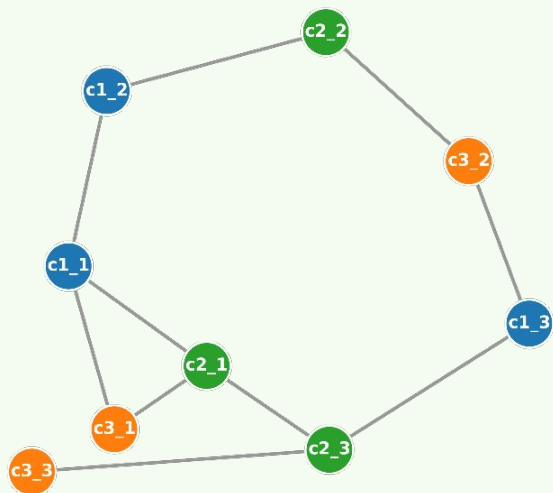


ReID Logic

NOTE: This is a dummy representation of the ReID graph. Not real IDs.

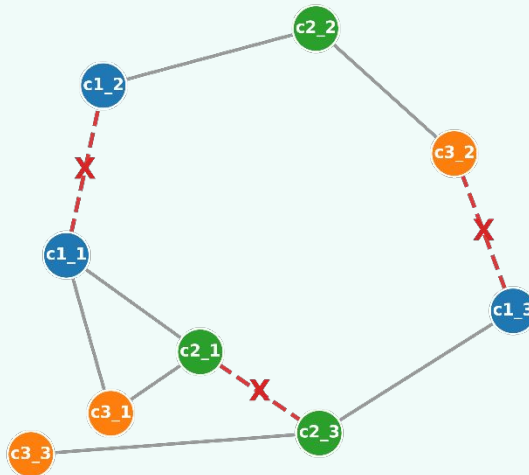
Bernat Medina

1. Graph Initialization (Raw ReID Visual Matches)



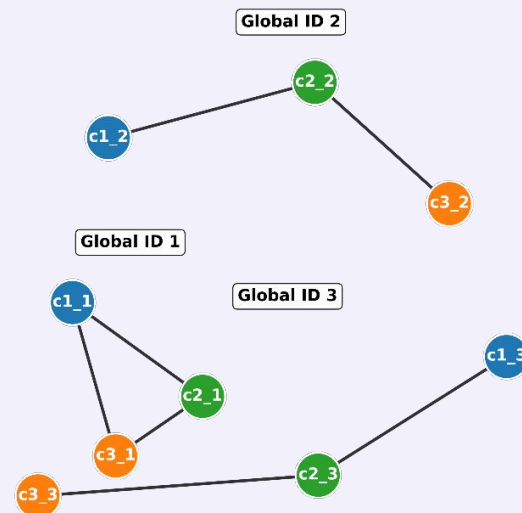
- **Nodes:** Global tracklets.
- **Edges:** Cosine embedding matches > 0.12.

2. Spatiotemporal Pruning (Physics Conflict Detection)



- Prune **Intra**-Camera conflicts.
- Prune **Inter**-Camera conflicts.

3. Greedy Clustering (Final Global IDs)

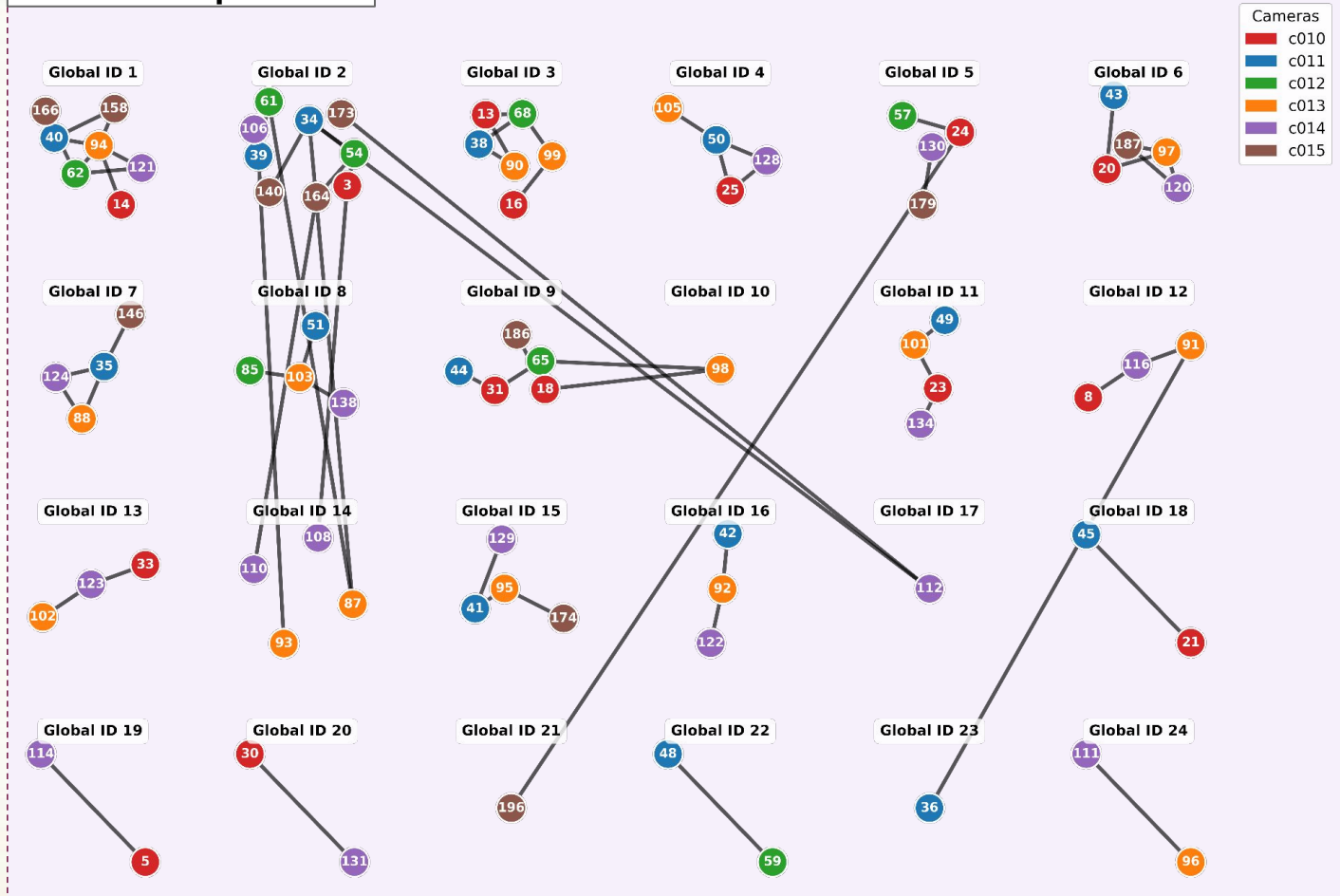


- Sort graph by node degree.
- Absorb neighboring nodes greedily.

ReID Logic

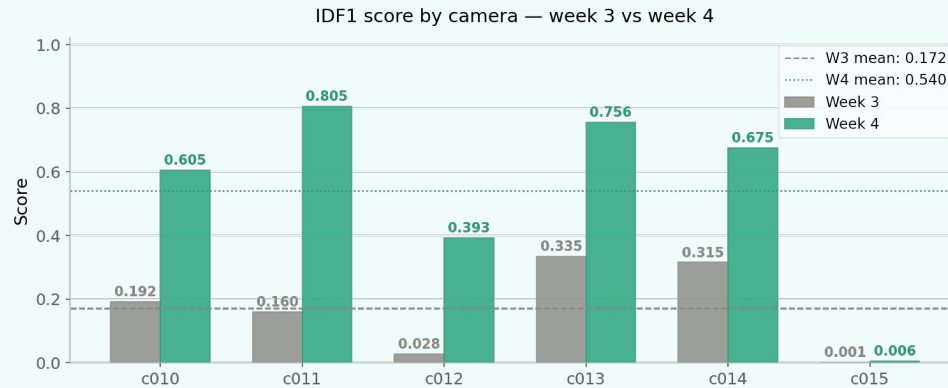
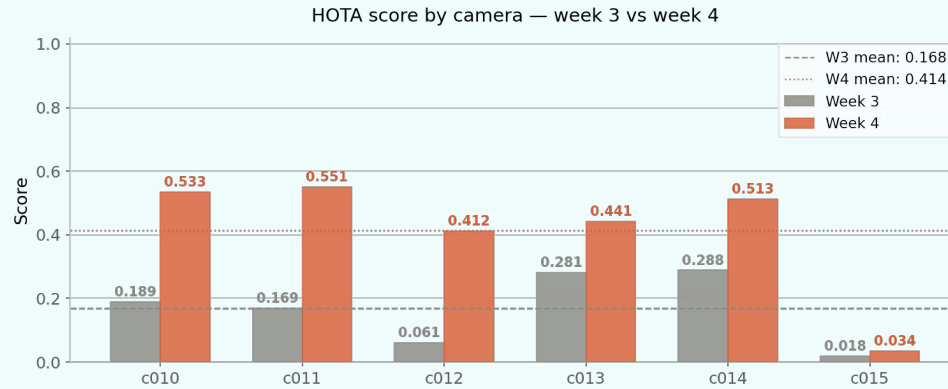
Global Graph ReID

Bernat Medina



Results

Bernat Medina



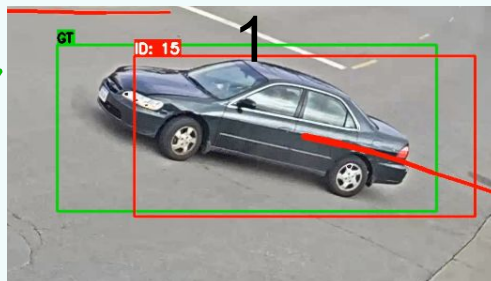
GLOBAL MTMC PERFORMANCE

HOTA	0.416
IDF1	0.569

Better results than last week!

Qualitative analysis

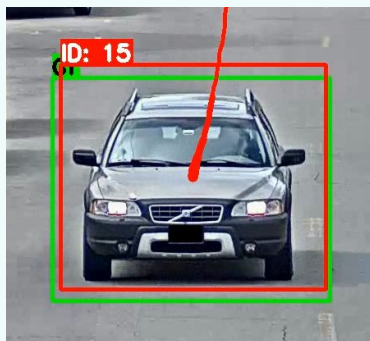
c01



c013



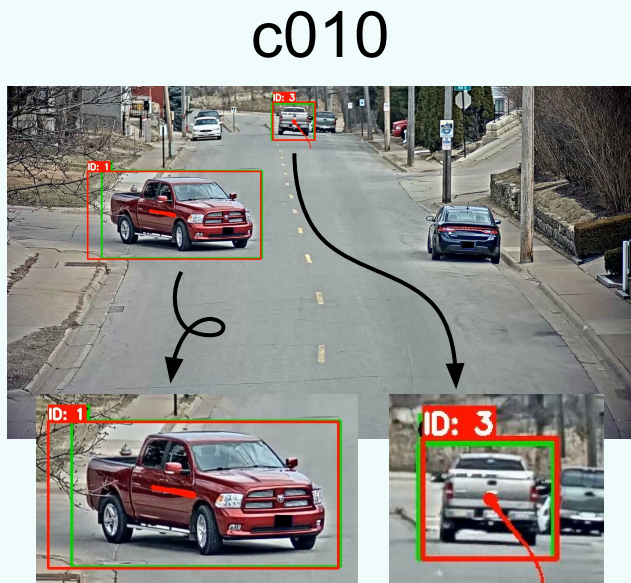
c014



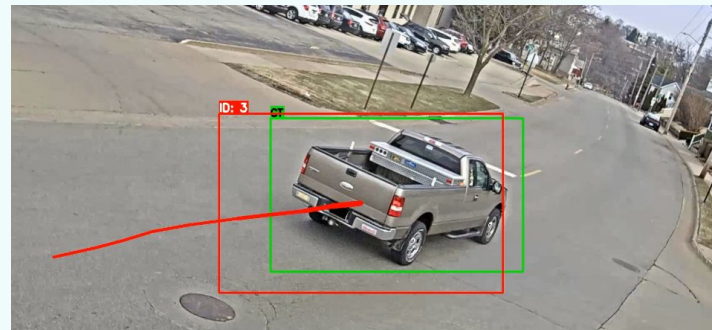
c015



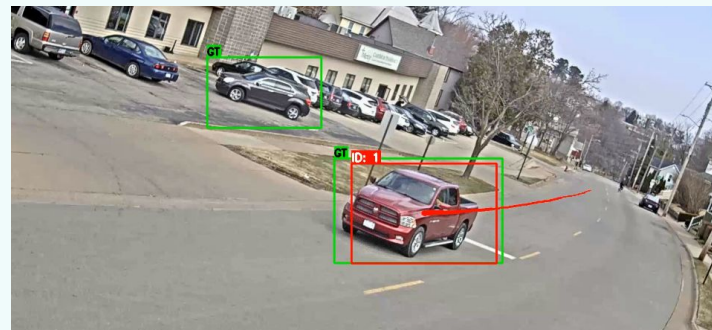
Qualitative analysis



c011

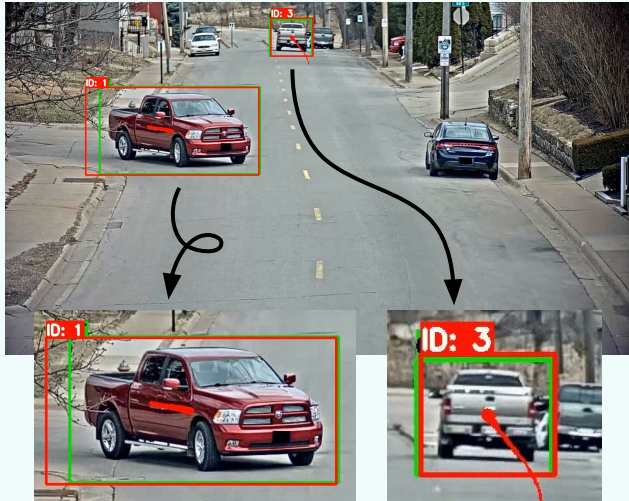


c011



Qualitative analysis

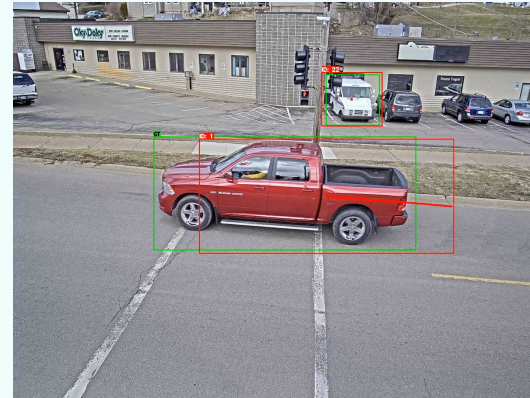
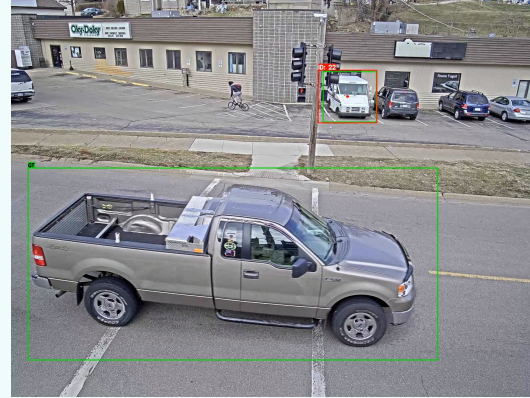
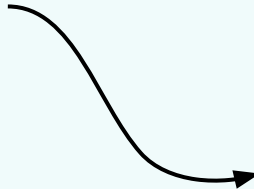
c010



c012

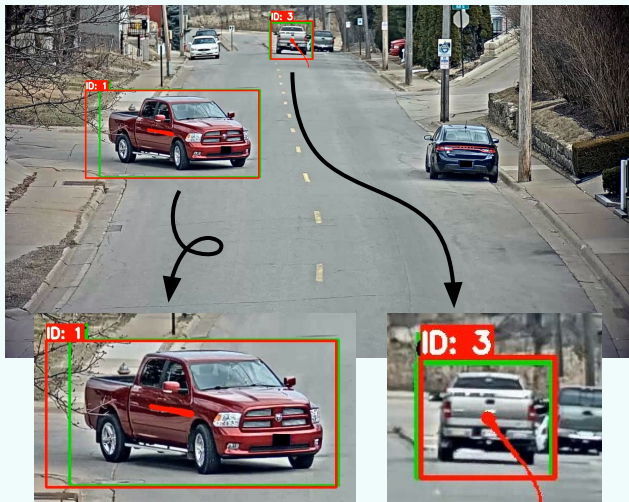


c012

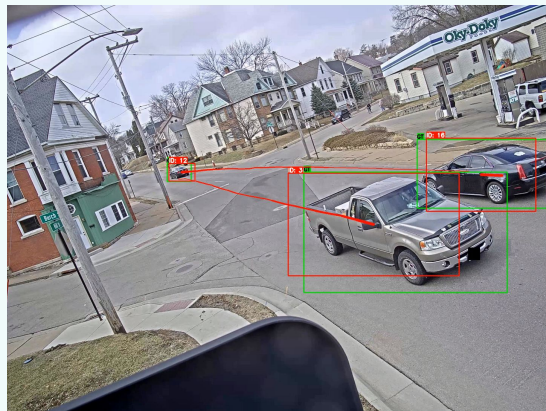


Qualitative analysis

c010



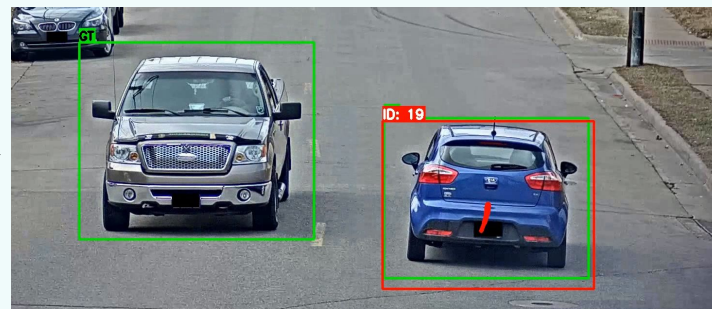
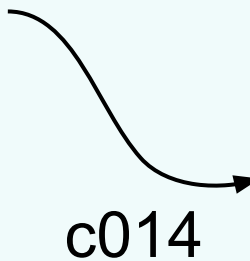
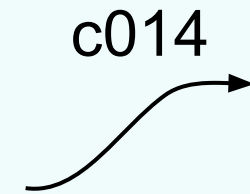
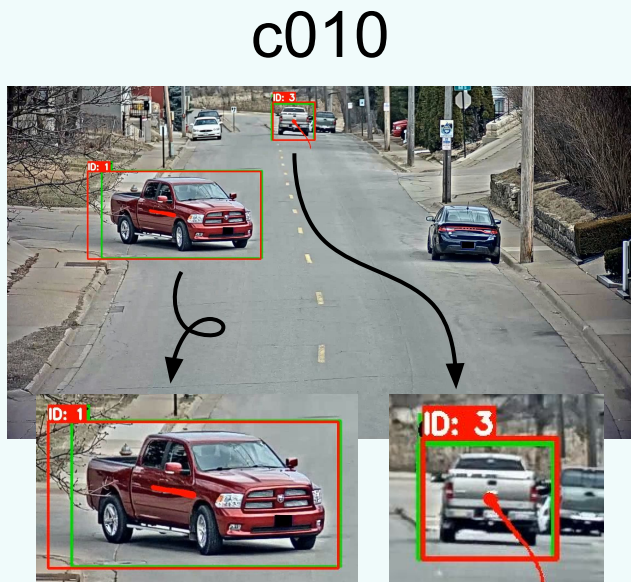
c013



c013



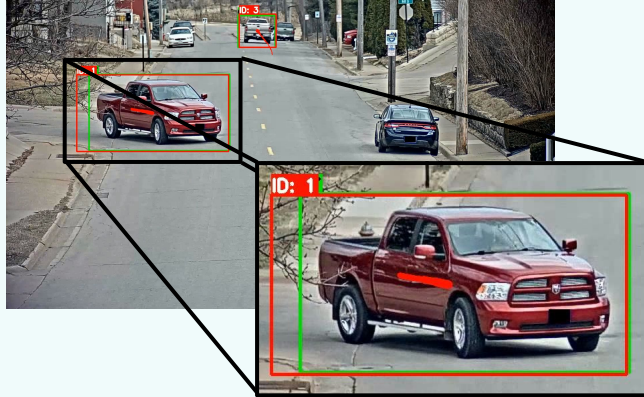
Qualitative analysis



Qualitative analysis

False Positive Example

c010



$\neq$

c015

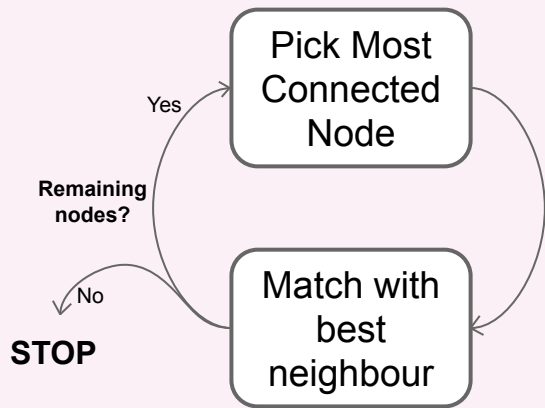


Conclusions

- ✦ **MTMC tracking approach.** The method combines tracking-by-detection: appearance-based re-ID embeddings, and graph-based global ID assignment across 6 cameras.
- ✦ **Offline pipeline.** The system processes all frames before making global ID decisions.
- ✦ **Significant improvement over week 3.** HOTA and IDF1 scores improved across all cameras, with the largest gains in c011 (+0.38 IDF1) and c014 (+0.37 IDF1).
- ✦ **Re-identification still incomplete.** Cross-camera matching recovers most vehicles but fragmented tracklets and similar-looking cars still cause missed links and duplicate IDs.

Future Work

Greedy approach

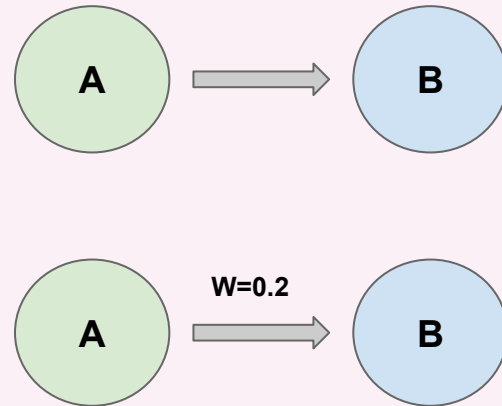


Camera quality filtering before matching



0.001 0.006
c015

Score-weighted graph edges



**THANKS FOR YOUR
ATTENTION**